PHY 338k Lab Report 4

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1 Introduction

Labs 9–11 of PHY 338k: Electronic Techniques explores

2 Lab 9: Digital Electronics I: Logic Levels and Gates

2.1 Input logic signals and output signal measurement

Include your measurements of the voltage thresholds and a couple of photos of your logical voltage source being read out on the LEDs for different combinations of zeros and ones

2.2 Quad NAND gate

Measure the output of the NAND gate for all four possible input states 00, 01, 10, and 11. Summarize your results with a truth table. Does it agree with what you expect?

Finally, measure and report the actual output voltages of the NAND gate for all four states. You should find a result that is very typical for CMOS logic gates.

Repeat the steps you just did. Do you still get the same truth table, experimentally?

Is the output voltage for the TTL version of the gate the same as the CMOS version, or different?

2.3 NAND gate circuits

2.3.1 NAND implementation of OR gate

Give your measured results in the form of a truth table.

2.3.2 A four input NAND gate circuit

Give your measured results in the form of a truth table.

Include a representative photo of the logic indicator output for one input example.

What logic function does this circuit carry out?

2.4 CMOS logic gates from individual transistors

2.4.1 Inverter with passive pull-up

Include a representative oscilloscope trace showing input and output at a low frequency and at a high frequency.

Give a qualitative description of the circuit behavior.

Also, measure the risetime and fall time of the circuit output and include the result in your report.

2.4.2 Inverter with active pull-up

Include a representative oscilloscope trace showing input and output at a low frequency and at a high frequency.

Give a qualitative description of the circuit behavior.

Also, measure the risetime and fall time of the circuit output and include the result in your report.

What are the main differences between the circuit with active pull-up and the circuit with passive pull- up. Explain why these differences occur. This provides one of the main reasons that essentially all logic circuits use active pull-up.

2.4.3 CMOS NAND gate constructed from individual transistors

Show the circuit diagram your used with just the CD4007 pins and your external connections, and also another diagram showing the complete circuit with the individual transistor connections.

Also show the CD4007 pins on the second diagram.

Show your result for the experimental truth table that you produced with your circuit.

3 Lab 10: Flip-Flops and Sequential Logic

3.1 Flip-flops from NAND gates

3.1.1 Asynchronous SR flip-flop

3.1.2 Clocked SR flip-flop

3.2 D-type flip-flop

3.2.1 D-type flip-flop operation

3.2.2 Flip-flop with feedback

3.2.3 Cascaded flip-flops

3.2.4 Shift register

4 Lab 11: Digital Circuits III

Conclusion

Labs 9-11

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